

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Data Structures & Algorithms**

Code No: **CACS201** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is data structure? Explain its importance.
3. Write an algorithm to convert infix expression to postfix.
4. Write a recursive program to generate Fibonacci number up to nth terms.
5. What is insertion sort? Trace and sort the following data using insertion sort: 90, 57, 80, 10, 22, 21, 45, 9, 78
6. What is hashing? Explain with example the collision resolution method open hashing.
7. Write the difference between serial and parallel algorithm with example.
8. Write a program to implement basic operation in queue.

Group C

Attempt any TWO questions[2x10=20]

9. What is circular linked list? Write a function to delete the node from linked list.
10. What do you mean by Huffman Algorithm? Explain with example. Construct the B tree of order 5 using following data: 20, 10, 26, 55, 80, 11, 9, 60, 67, 55, 22, 76, 56, 45, 34, 100, 150
11. What do you mean by MST? Explain Kruskal's algorithm with example.